

Course Code:B23CE4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
GEOTECHNICAL ENGINEERING II					
Civil Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define subsoil exploration and state its objectives.	1	1	2
	b).	Differentiate between disturbed and undisturbed samples	1	2	2
	c).	What is safe bearing capacity of soil?	2	1	2
	d).	Explain the difference between immediate settlement and consolidation settlement.	2	2	2
	e).	Define pile foundation and mention its types.	3	1	2
	f).	Distinguish between end-bearing piles and friction piles.	3	2	2
	g).	What is meant by factor of safety in slope stability?	4	1	2
	h).	Explain the concept of a critical slip surface.	4	2	2
	i).	Define active earth pressure.	5	1	2
	j).	Differentiate between Rankine's theory and Coulomb's theory of earth pressure.	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain two boring methods with neat sketch	1	2	5
	b).	The measured N-value in a borehole is found to be 20.The depth of the bore hole was 15m.The watertable was located 1.5mbelow ground level. The soil is fine silty and its unit weight above and below water table was 16 and 21kN/m ³ respectively. Calculate the corrected value of N.	1	3	5
OR					
3.	a).	Discuss standard penetration test. What are the various corrections?	1	2	5
	b).	Compute the area ratio of a thin-walled tube samples having an external diameter of 8 cm and a wall thickness of 3 mm. Do you recommend the sampler for obtaining undisturbed soil samples?	1	3	5
UNIT-2					
4.	a).	What are different types of shallow foundation? Explain with the help of sketches.	2	2	5
	b).	A Square footing is to be designed to carry a total load of 500KN in a soil having C=10 kN/m ² , Ø=25 ⁰ , γ=16kN/m ³ .The footing is to be laid at a depth of 2 m below ground level and F.S is 3. Find the Dimension of the footing. Given N _c =22, N _q =11, N _γ =9.	2	3	5

		OR			
5.	a).	Derive the expression for Terzaghi's Bearing Capacity theory and write the assumptions?	2	2	5
	b).	A footing 4m x 2m in plan, transmits a pressure of 150 N/sq.m on a cohesive soil having $E = 6 \times 10^4$ kN/sq.m and $\mu = 0.50$. Determine the immediate settlement of the footing at the centre, assuming it to be (a) a flexible footing, $I = 1.32$ and (b) a rigid footing, $I = 1.20$	2	3	5
		UNIT-3			
6.	a).	Explain Pneumatic Caisson with neat sketch and mention its advantages and disadvantages.	3	2	5
	b).	A concrete pile, 25cm in diameter is driven in to a dense sand ($\phi = 30^\circ$, $\gamma = 20$ kN/m ³ , $K = 1$, $\tan \delta = 0.70$) for a depth of 6m. If the water table is at 2m from the ground level, estimate the safe load, taking Factor of safety 3.	3	3	5
		OR			
7.	a).	Discuss the Causes and Remedies for Tilts and Shifts (with neat sketches) in Well Foundations.	3	2	5
	b).	A pile group consists of 9 piles of 300 mm dia and length 10m, is arranged in 3 rows at spacing of 750 mm c/c. The piles are driven into a deposit of clay whose properties are $C = 100$ kN/m ² , $\gamma = 20$ kN/m ³ . Determine the safe load on the pile group. Take F.S=3, $\alpha = 0.6$, $N_c = 9$	3	3	5
		UNIT-4			
8.	a).	Discuss the Friction circle method for the stability analysis of slopes. Can this method be used for purely cohesive soil?	4	2	5
	b).	A cut of length 10m is made in a cohesive soil deposit ($C = 30$ kN/m ² , $\phi = 0^\circ$, $\gamma = 20$ kN/m ³). There is a hard stratum under the cohesive soil at a depth of 12m below the original ground surface. If the required factor of safety is 1.5, determine the safe slope.	4	3	5
		OR			
9.	a).	Explain types of slope failures	4	2	5
	b).	A slope is to be Construction in a soil For which $C = 0$ and $\phi = 36^\circ$. It is to be assumed that the water level may occasionally reach the surface of a slope, with seepage taking place parallel to the slope. Determine the maximum slope angle for a factor of safety 1.5, assuming a potential failure surface parallel to the slope What would be the factor of safety of the slope, constructed at this angle, if the water tables should be well below the surface?	4	3	5
		UNIT-5			
10.	a).	What are the assumptions of Rankine's theory? Derive the expressions for active pressure and passive pressure	5	2	5
	b).	Determine the passive pressure by Rankine's theory per unit run for a retaining wall 4m high, with $i = 15^\circ$, $\phi = 30^\circ$ and $\gamma = 20$ kN/m ³ . The	5	3	5

		back face of the wall is smooth and vertical.			
		OR			
11.	a).	Discuss Culmann's method for the determination of active earth pressure.	5	2	5
	b).	A 5m high rigid retaining wall has to retain a backfill of dry cohesion less soil having the following properties: $\phi = 30^\circ$, $e=0.74$ and $G=2.68$. Plot the distribution of Rankine lateral earth pressure on the wall and determine the magnitude and point of application of the resultant thrust.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
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Course Code:B23HS4103																			
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23														
IV B.Tech. I Semester MODEL QUESTION PAPER																			
ADVANCED CONSTRUCTION MANAGEMENT																			
Civil Engineering																			
Time: 3 Hrs.			Max. Marks: 70 M																
Answer Question No.1 compulsorily																			
Answer ONE Question from EACH UNIT																			
Assume suitable data if necessary																			
10 x 2 = 20 Marks																			
			CO	KL	M														
1.	a).	Define Bar Chart and mention its limitations.	1	2	2														
	b).	What is a Dummy Activity in CPM/PERT network?	1	2	2														
	c).	Define Earliest Occurrence Time.	2	2	2														
	d).	Differentiate between Total Float and Free Float.	2	2	2														
	e).	Define Cost Slope in project crashing.	3	2	2														
	f).	What are Direct and Indirect Costs?	3	2	2														
	g).	List any four objectives of Quality Management in construction.	4	2	2														
	h).	State any two safety measures followed at construction sites.	4	2	2														
	i).	What is BIM?	5	2	2														
	j).	Mention any two advantages of 3D modelling in construction projects.	5	2	2														
5 x 10 = 50 Marks																			
UNIT-1																			
2.	a).	Explain the procedure for drawing a CPM network with suitable example. Discuss the rules for network construction.	1	3	5														
	b).	Compare PERT and CPM techniques with suitable applications in construction projects.	1	3	5														
OR																			
3.	a).	Explain Bar Charts and Milestone Charts used in project planning. Mention their merits and demerits.	1	2	5														
	b	Construct a project network for the following activities and number the events using Fulkerson’s Rule.	1	3	5														
<table><tr><th>Activity</th><th>Predecessor</th></tr><tr><td>A</td><td>–</td></tr><tr><td>B</td><td>A</td></tr><tr><td>C</td><td>A</td></tr><tr><td>D</td><td>B</td></tr><tr><td>E</td><td>B,C</td></tr><tr><td>F</td><td>D,E</td></tr></table>			Activity	Predecessor	A	–	B	A	C	A	D	B	E	B,C	F	D,E			
Activity	Predecessor																		
A	–																		
B	A																		
C	A																		
D	B																		
E	B,C																		
F	D,E																		

		UNIT-2																								
4.	a).	Explain the concepts of Earliest Start Time, Latest Start Time, Earliest Finish Time, and Latest Finish Time in network analysis.	2	3	5																					
	b).	A project consists of the following activities: <table border="1"><thead><tr><th>Activity</th><th>Duration (Days)</th><th>Predecessor</th></tr></thead><tbody><tr><td>A</td><td>4</td><td>–</td></tr><tr><td>B</td><td>6</td><td>A</td></tr><tr><td>C</td><td>5</td><td>A</td></tr><tr><td>D</td><td>3</td><td>B</td></tr><tr><td>E</td><td>2</td><td>C</td></tr><tr><td>F</td><td>4</td><td>D,E</td></tr></tbody></table> Draw the network and determine the critical path and project duration.	Activity	Duration (Days)	Predecessor	A	4	–	B	6	A	C	5	A	D	3	B	E	2	C	F	4	D,E	2	3	5
Activity	Duration (Days)	Predecessor																								
A	4	–																								
B	6	A																								
C	5	A																								
D	3	B																								
E	2	C																								
F	4	D,E																								
		OR																								
5.	a).	Define Slack and Floats. Explain different types of floats used in CPM networks.	2	2	5																					
	b).	For a given network, calculate Total Float and Free Float for each activity and identify critical activities.	2	4	5																					
		UNIT-3																								
6.	a).	Explain Cost–Time Optimization in construction projects.	3	3	5																					
	b).	An activity has the following data: Normal Time = 10 days Crash Time = 6 days Normal Cost = ₹20,000 Crash Cost = ₹32,000 Calculate the Cost Slope.	3	3	5																					
		OR																								
7.	a).	Discuss Project Budgeting and Cost–Benefit Analysis in construction management.	3	3	5																					
	b).	Explain project updating and crashing techniques with suitable examples.	3	4	5																					
		UNIT-4																								
8.	a).	Explain Quality Management practices adopted in construction projects.	4	3	5																					
	b).	Discuss the importance of Material Management and Machinery Management in construction projects.	4	3	5																					
		OR																								
9.	a).	Describe various On-site Safety Management procedures followed in construction industries.	4	3	5																					
	b).	Explain preventive measures to reduce accidents at construction sites.	4	3	5																					

		UNIT-5			
10.	a).	Explain the role of Building Information Modelling (BIM) in project management.	5	3	5
	b).	Discuss the advantages of collaboration and efficiency improvement using BIM tools.	5	3	5
		OR			
11.	a).	Explain various 3D modelling applications used in modern construction management.	5	3	5
	b).	Compare traditional construction management methods with BIM-based project management systems.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



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Course Code:B23CE4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRESTRESSED CONCRETE					
Civil Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a	Why are high-strength concrete and high-strength steel used in prestressed concrete?	1	1	2
	b	What are the advantages of post-tensioning over pre-tensioning?	1	2	2
	c	What are the basic assumptions in prestressed concrete analysis?	2	2	2
	d	Define kern distance and its significance in prestressed concrete?	2	2	2
	e	What is the difference between cracked and uncracked sections?	3	2	2
	f	What are the types of flexural failure in prestressed concrete?	3	2	2
	g	What factors affect transmission length?	4	2	2
	h	What are the assumptions and limitations in Guyon's method?	4	2	2
	i	Define concordant tendon profile.	5	2	2
	j	What is short-term and long-term deflection?	5	2	2
5 x 10 = 50 Marks					
UNIT-I					
2.	a).	Describe the materials used in prestressed concrete, including high-strength concrete and high-strength steel.	1	2	5
	b).	Explain pre-tensioning and post-tensioning methods.	1	2	5
OR					
3.	a).	Mention the advantages and limitations of prestressed concrete?	1	2	5
	b).	Explain the Applications of Prestressed concrete.	1	2	5
UNIT-II					
4.		A rectangular concrete beam of cross-section 300 mm deep and 200 mm wide is prestressed by means of 15 wires of 5 mm diameter located 65 mm from the bottom of the beam and 3 wires of diameter of 5 mm, 25 mm from the top. Assuming the prestress in the steel as 840 N/mm ² , calculate the stresses at the extreme fibres of the mid-span section when the beam is supporting its own weight over a span of 6 m. If a uniformly distributed live load of 6 kN/m is imposed, evaluate the maximum work-ing stress in concrete. The density of concrete is 24 kN/m ³	2	4	10
OR					
5.		A pretensioned beam 250 mm wide and 300 mm deep is prestressed by 12 wires, each 7 mm diameter, initially stressed to 1200 N/mm ² with their centroids located 100 mm from the soffit. Estimate the final percentage loss of stress due to elastic deformation, creep, shrinkage and relaxation	2	4	10

		using IS: 1343 code and the following data: Relaxation of steel stress = 90 N/mm^2 $E_c = 35 \text{ kN/mm}^2$, $E_s = 210 \text{ kN/mm}^2$ Creep coefficient (ϕ) = 1.6 Residual shrinkage strain = 3×10^{-4}			
		UNIT-III			
6.		A pretensioned, T-section has a flange which is 300 mm wide, 200 mm thick. The rib is 150 mm wide by 350 mm deep. The effective depth of the cross-section is 500 mm. Given, $A_p = 200 \text{ mm}^2$, $f_{ck} = 50 \text{ N/mm}^2$ and $f_{t1} = 1600 \text{ N/mm}^2$, estimate the ultimate moment capacity of the T-section using the Indian Standard Code regulations.	3	4	10
		OR			
7.	a).	Explain modes of shear failures in prestressed concrete beams?	3	2	2
	b).	The support section of a prestressed concrete beam, 120 mm wide and 250 mm deep, is required to support an ultimate shear force of 60 kN. The compressive prestress at the centroidal axis is 5 N/mm^2 . The characteristic cube strength of concrete is 40 N/mm^2 . The cover to be tension reinforcement is 50 mm. If the characteristic tensile strength of steel in stirrups is 250 N/mm^2 , design suitable reinforcement at the section using IS: 1343 code specifications.	3	4	8
		UNIT-IV			
8.		The end block of a post-tensioned prestressed member is 550 mm wide and 550 mm deep. Four cables, each made up of 7 wires of 12 mm diameter strands and carrying a force of 1000 kN, are anchored by plate anchorages, 150 mm $\times$ 150 mm, located with their centres at 125 mm from the edges of the end block. The cable duct is of 50 mm diameter. The 28 days cube strength of concrete f_{cu} is 45 N/mm^2 . The cube strength of concrete at transfer f_{ci} is 25 N/mm^2 . Permissible bearing stresses behind anchorages should conform with IS 1343. The characteristic yield stress in mild steel anchorage reinforcement is 260 N/mm^2 . Design suitable anchorage for the end block	4	4	10
		OR			
9.		A Freyssinet anchorage (125 mm diameter), carrying 12 wires of 7 mm diameter stressed to 950 N/mm^2 , is embedded concentrically in the web of an I-section beam at the ends. The thickness of the web is 225 mm. Evaluate the maximum tensile stress and the bursting tensile force in the end block. Design the reinforcement for the end block?	4	4	10
		UNIT-V			
10.		A prestressed concrete beam having a cross-sectional area (A) of $5 \times 10^4 \text{ mm}^2$ is simply supported over a span of 10 m. It supports a uniformly distributed imposed load of 3 kN/m, half of which is non-permanent. The tendon follows a trapezoidal profile with an eccentricity of 100 mm within the middle-third of the span and varies linearly from the third-span points to zero at the supports. The area of tendons $A_p = 350 \text{ mm}^2$ have effective prestress of 1290 N/mm^2 immediately after transfer. Using the following data, calculate	5	4	10

		1. the short-term deflections, and 2. the long-term deflections. Assume $I_g = 4.5 \times 10^8 \text{ mm}^4$ $A = 5 \times 10^4 \text{ mm}^2$ $E_c = 34 \text{ kN/mm}^2$ $E_s = 200 \text{ kN/mm}^2$ Density of concrete = 23.6 kN/m^3) Creep coefficient = 2, Concrete shrinkage $\epsilon_{cs} = 450 \times 10^{-6}$ Relaxation of steel stress = 10%			
		OR			
11.		A two-span continuous prestressed concrete beam ABC (AB = BC = 15m) has a uniform cross-section with a width of 250 mm and a depth of 600 mm. The cable carrying an effective prestressing force of 500kN is parallel to the axis of the beam and located at an eccentricity of 200 mm. (a) Determine the secondary and resultant moment development at mid support section B (b) If the beam supports an imposed load of 2.4 kN/m , calculate the resultant stresses developed at the top and bottom of the beam at B. Also calculate the resultant line of thrust through the beam AB.	5	4	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



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Course Code: B23CE4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ADVANCED ENVIRONMENTAL ENGINEERING					
Civil Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define self-purification in streams.	1	1	2
	b).	Identify the factors affecting dissolved oxygen balance.	1	1	2
	c).	State the principle of nitrification.	2	1	2
	d).	Select a suitable biological treatment method for nitrogen removal.	2	3	2
	e).	List any two characteristics of dairy wastewater.	3	1	2
	f).	Select a suitable treatment method for sugar industry wastewater.	3	3	2
	g).	Define solid waste classification.	4	1	2
	h).	Identify types of air pollutants with examples.	4	2	2
	i).	State any two effects of air pollution on human health.	5	1	2
	j).	Select a suitable device for control of particulate matter.	5	3	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Determine the dissolved oxygen deficit at a downstream point using the Streeter–Phelps equation for given data.	1	3	5
	b).	Examine the impact of various pollutants on stream water quality and aquatic life.	1	4	5
		OR			
3.	a).	Determine the critical deficit and critical time for a stream using appropriate equations.	1	3	5
	b).	Compare the factors affecting self-purification of streams under different environmental conditions.	1	4	5
		UNIT-2			
4.	a).	Use principles of nitrification and denitrification in biological wastewater treatment with suitable reactions.	2	3	5
	b).	Examine the working mechanism and performance of Rotating Biological Contactors (RBC).	2	4	5
		OR			
5.	a).	Determine phosphate removal in activated sludge process using suitable treatment steps.	2	3	5
	b).	Compare anaerobic filters and U-tube aeration systems in terms of efficiency and application.	2	4	5
		UNIT-3			

6.	a).	Select suitable treatment methods for sugar industry wastewater based on its characteristics.	3	3	5
	b).	Examine the characteristics of dairy wastewater and associated treatment challenges.	3	4	5
		OR			
7.	a).	Select appropriate treatment methods for pulp and paper industry wastewater.	3	3	5
	b).	Compare different industrial wastewater treatment approaches with respect to efficiency and suitability.	3	4	5
		UNIT-4			
8.	a).	Use solid waste management principles for collection and transportation of municipal waste.	4	3	5
	b).	Examine treatment methods such as composting and sanitary landfill with respect to their effectiveness.	4	4	5
		OR			
9.	a).	Classify air pollutants based on their sources and characteristics.	4	3	5
	b).	Examine the influence of meteorological parameters on air pollution dispersion.	4	4	5
		UNIT-5			
10.	a).	Select suitable air pollution control equipment for given particulate conditions.	5	3	5
	b).	Examine the effects of air pollution on human health, plants, and environment.	5	4	5
		OR			
11.	a).	Select appropriate noise control techniques for different environmental conditions.	5	3	5
	b).	Examine the impact of noise pollution on human health and surroundings.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23CE4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
URBAN HYDROLOGY					
Civil Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define urban hydrologic cycle.	1	1	2
	b).	What is urbanisation?	1	1	2
	c).	What is time of concentration?	2	1	2
	d).	Define IDF curve.	2	1	2
	e).	What is rational method?	3	1	2
	f).	What is runoff coefficient?	3	1	2
	g).	What are Best Management Practices (BMPs)?	4	1	2
	h).	Define detention and retention.	4	1	2
	i).	What is stormwater drainage system?	5	1	2
	j).	What is master drainage plan?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain urban hydrologic cycle and effects of urbanisation on hydrology.	1	2	5
	b).	Discuss trends in urbanisation and their impact on water cycle.	1	2	5
		OR			
3.	a).	Explain how urbanisation alters infiltration, runoff, and evapotranspiration.	1	2	5
	b).	Discuss the importance of studying urban hydrology in modern cities.	1	2	5
		UNIT-2			
4.	a).	Explain Intensity-Duration-Frequency (IDF) curves and their significance.	2	2	5
	b).	Discuss methods of estimating time of concentration.	2	2	5
		OR			
5.	a).	Explain design storms and their role in urban drainage design.	2	2	5
	b).	Discuss the importance of short-duration rainfall data in urban hydrology.	2	2	5
		UNIT-3			
6.	a).	Explain rational method for peak flow estimation with assumptions.	3	2	5
	b).	Describe NRCS curve number method.	3	2	5

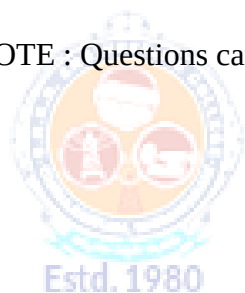
		OR			
7.	a).	Explain components of urban drainage systems such as open channels and underground drains.	3	2	5
	b).	Discuss wastewater and stormwater reuse in urban areas.	3	2	5
		UNIT-4			
8.	a).	Explain stormwater drainage structures and network design.	4	2	5
	b).	Discuss Best Management Practices (BMPs) in urban flooding control.	4	2	5
		OR			
9.	a).	Explain detention and retention facilities with advantages.	4	2	5
	b).	Discuss models used for stormwater management.	4	2	5
		UNIT-5			
10.	a).	Explain the concept of master drainage plan and its objectives.	5	2	5
	b).	Discuss the relationship between urban planning and drainage systems.	5	2	5
		OR			
11.	a).	Explain the steps involved in preparing a master drainage plan.	5	2	5
	b).	Discuss the role of models in urban drainage planning.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



Course Code:B23CE4105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
FOUNDATIONS ON EXPANSIVE SOILS					
Civil Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define cation exchange in clay minerals.	1	1	2
	b).	Explain the concept of soil–water interaction in expansive soils.	1	2	2
	c).	What is swelling pressure?	2	1	2
	d).	Explain the factors affecting swell potential of expansive soils.	2	2	2
	e).	What is an under-reamed pile?	3	1	2
	f).	Explain the purpose of a sand cushion in expansive soils.	3	2	2
	g).	What is soil stabilization?	4	1	2
	h).	Explain the role of lime in stabilizing expansive soils.	4	2	2
	i).	Define matrix suction.	5	1	2
	j).	Explain Katti’s concept of bilinear strength envelope.	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain the structure of clay minerals and their influence on the behavior of expansive soils.	1	2	5
	b).	Explain soil–water interaction and its role in volume change behavior of expansive soils.	1	2	5
OR					
3.	a).	Explain the nature of expansive soils and the role of clay mineralogy in their engineering behavior.	1	2	5
	b).	Explain the concept of cation exchange and its effect on soil properties.	1	2	5
UNIT-2					
4.	a).	Explain the factors affecting swelling pressure of expansive soils.	2	2	5
	b).	In a laboratory test, a 15 mm thick soil sample swells by 1.2 mm under no load. Under a load of 60 kPa, swell reduces to 0.4 mm. Determine: (a) Free swell percentage (b) Swelling pressure (approximate)	2	3	5
OR					
5.	a).	Explain different methods used for determination of swell potential and heave.	2	2	5
	b).	A 3 m thick expansive clay layer has a swell potential of 4%. Estimate	2	3	5

		the total heave. If a surcharge reduces swell to 1.5%, determine the reduced heave.			
		UNIT-3			
6.	a).	Explain the construction procedure of under-reamed pile foundations.	3	2	5
	b).	An under-reamed pile has a shaft diameter of 0.3 m and bulb diameter of 0.75 m, embedded to a depth of 7 m in clay with cohesion 25 kN/m ² . Calculate the ultimate load capacity considering shaft and end bearing.	3	3	5
		OR			
7.	a).	Explain the function of sand cushion and CNS layer technique in expansive soils.	3	2	5
	b).	A group of 9 under-reamed piles (3×3 arrangement) is installed in clay. Each pile has a diameter of 0.35 m, length of 8 m, and cohesion of soil is 28 kN/m ² . Assume adhesion factor = 0.6 and N _c = 9. Determine: (a) Ultimate load capacity of a single pile (consider both shaft resistance and end bearing) (b) Group capacity if efficiency = 0.8 (c) Safe load for the pile group with a factor of safety of 2.5	3	3	5
		UNIT-4			
8.	a).	Explain the mechanism of lime stabilization in expansive soils.	4	2	5
	b).	A soil stabilized with lime shows an increase in UCS from 120 kPa to 300 kPa. Determine: (a) Percentage increase in strength (b) Improvement factor	4	3	5
		OR			
9.	a).	Explain vertical and horizontal moisture barrier techniques.	4	2	5
	b).	Lime columns of 0.25 m diameter are installed at 1 m spacing in a square pattern. Determine: (a) Area replacement ratio (b) Expected strength improvement if directly proportional	4	3	5
		UNIT-5			
10.	a).	Explain Katti's concept of bilinear strength envelope.	5	2	5
	b).	Using Fredlund's shear strength equation: $\tau = c' + (\sigma - u_a)\tan \phi' + (u_a - u_w)\tan \phi^b$ Derive an expression for shear strength of soil when it becomes fully saturated, and explain the significance of each term.	5	3	5
		OR			
11.	a).	Explain the axis translation technique for measurement of matrix suction.	5	2	5
	b).	A partly saturated soil has the following properties:	5	3	5

	$c' = 15 \text{ kPa}, \phi' = 28^\circ, \phi^b = 12^\circ$. If the net normal stress ($\sigma - u_a$) is 120 kPa and matric suction ($u_a - u_w$) is 50 kPa: Determine: (a) Shear strength of soil (b) Percentage contribution of matric suction to total shear strength			
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CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code:B23CE4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
GEOSYNTHETICS & ITS APPLICATIONS					
Civil Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define geotextiles and geogrids	1	1	2
	b).	Explain the difference between woven and non-woven geotextiles.	1	2	2
	c).	What are geomembranes?	2	1	2
	d).	Explain the functions of geocomposites.	2	2	2
	e).	What is base reinforcement in pavements?	3	1	2
	f).	Explain the components of a reinforced soil retaining wall.	3	2	2
	g).	What is a landfill liner?	4	1	2
	h).	Explain the use of geomembranes in canal lining.	4	2	2
	i).	What is CQA in geosynthetics?	5	1	2
	j).	Explain the importance of proper installation of geosynthetics.	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain the types and manufacturing methods of geotextiles.	1	2	5
	b).	Describe the tests to determine the tensile strength of geogrids.	1	2	5
OR					
3.	a).	Describe the different types of geogrids and their characteristics.	1	2	5
	b).	Explain the physical, mechanical and hydraulic properties of geotextiles.	1	2	5
UNIT-2					
4.	a).	Explain the types, materials and manufacturing methods of geomembranes.	2	2	5
	b).	Explain the structure of geonets and discuss their functions in drainage applications.	2	2	5
OR					
5.	a).	Explain the types and functions of geocomposites with examples.	2	2	5
	b).	Describe geosynthetic clay liners and explain their basic properties and functions.	2	2	5
UNIT-3					

6.	a).	Explain the role of geosynthetics in base reinforcement and load distribution in pavements.	3	2	5
	b).	A reinforced soil retaining wall of height 5 m is proposed with panel facing. The unit weight of soil is 19 kN/m ³ , angle of internal friction is 32°, and a surcharge load of 10 kN/m ² is acting on the backfill. Apply design concepts to: (i) Calculate the total active earth pressure acting on the wall (ii) Determine the factor of safety against sliding if the base friction coefficient is 0.5 (iii) Comment on the adequacy of the design (Assume Rankine conditions and neglect cohesion)	3	3	5
		OR			
7.	a).	Describe the components and construction methods of reinforced soil retaining walls.	3	2	5
	b).	A reinforced soil retaining wall of height 6 m is constructed using geogrids as reinforcement. The unit weight of backfill soil is 18 kN/m ³ and the angle of internal friction is 30°. Using Rankine's theory, determine: (i) The active earth pressure at the base (ii) The required minimum length of reinforcement assuming it is 0.7H (iii) Suggest suitable vertical spacing of reinforcement layers (Assume level backfill and no surcharge)	3	3	5
		UNIT-4			
8.	a).	Explain the use of geomembranes in lining systems for reservoirs and canals.	4	2	5
	b).	Describe the components and functions of a landfill liner system.	4	2	5
		OR			
9.	a).	Explain the role of geosynthetics in erosion control using mats and geocells.	4	2	5
	b).	Describe the use of geosynthetics in coastal and riverbank protection.	4	2	5
		UNIT-5			
10.	a).	Explain performance and prescriptive specifications used in geosynthetics.	5	2	5
	b).	Describe the installation procedures of geosynthetics including handling, seaming and anchoring.	5	2	5
		OR			
11.	a).	Explain the methods of seaming and seam testing for geomembranes.	5	2	5
	b).	Describe Construction Quality Assurance (CQA), Quality Control (CQC) and post-construction monitoring in geosynthetics.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CE4108					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ENVIRONMENTAL IMPACT ASSESSMENT					
Civil Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Environmental Impact Assessment (EIA).	1	1	2
	b).	List the stages involved in EIA process.	1	1	2
	c).	State any two EIA methodologies.	2	1	2
	d).	Select a suitable EIA method for assessing groundwater impact.	2	3	2
	e).	List any two parameters for surface water quality assessment.	3	1	2
	f).	Apply a suitable mitigation measure for air pollution impact.	3	3	2
	g).	Define environmental risk assessment.	4	1	2
	h).	Identify impacts of deforestation on wildlife.	4	2	2
	i).	State the purpose of EIA notification.	5	1	2
	j).	Select a step involved in environmental audit process.	5	3	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Use GIS and remote sensing data to prepare an environmental baseline map for a proposed project site, highlighting key environmental parameters.	1	3	5
	b).	Justify the role of stakeholders and cost–benefit analysis in environmental decision-making with suitable examples.	1	4	5
		OR			
3.	a).	Demonstrate the stages of EIA by evaluating a proposed development project systematically.	1	3	5
	b).	Compare Environmental Impact Statement (EIS) and Environmental Management Plan (EMP) with respect to their functions in project evaluation.	1	4	5
		UNIT-II			
4.	a).	Demonstrate matrix and network methods for identifying environmental impacts of a development project.	2	3	5
	b).	Classify different EIA methodologies based on their suitability for land use impact assessment.	2	4	5
		OR			
5.	a).	Use the overlay method to assess environmental impacts for a selected project site.	2	3	5

	b).	Infer the effects of developmental activities on soil and groundwater quality using relevant parameters.	2	4	5
		UNIT-III			
6.	a).	Demonstrate impact prediction techniques for assessing surface water quality.	3	3	5
	b).	Justify the effectiveness of mitigation measures adopted for controlling air pollution.	3	4	5
		OR			
7.	a).	Use methodology for assessing impacts on the biological environment.	3	3	5
	b).	Categorize the significance of impact assessment results for planning mitigation measures.	3	4	5
		UNIT-IV			
8.	a).	Explain the steps involved in environmental risk assessment for a given developmental activity.	4	2	5
	b).	Infer the impacts of deforestation on vegetation and wildlife.	4	4	5
		OR			
9.	a).	Use appropriate risk management strategies to minimize environmental impacts in EIA studies.	4	3	5
	b).	Classify the uncertainties involved in environmental risk assessment.	4	4	5
		UNIT-V			
10.	a).	Explain the procedure for obtaining environmental clearance as per EIA notification guidelines.	5	3	5
	b).	Justify the role of environmental audit in ensuring compliance with environmental regulations.	5	4	5
		OR			
11.	a).	Prepare an Environmental Impact Assessment (EIA) report by outlining its essential components.	5	3	5
	b).	Compare the methods used in evaluation of EIA reports and environmental audit data.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23CE4109					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
DISASTER MANAGEMENT AND PREPAREDNESS					
Civil Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define hazard and vulnerability.	1	1	2
	b).	What is disaster resilience?	1	1	2
	c).	Mention two disaster-prone regions in India.	2	1	2
	d).	What is seismic zoning?	2	1	2
	e).	Define disaster risk reduction (DRR).	3	1	2
	f).	What is vulnerability assessment?	3	1	2
	g).	List the phases of disaster management cycle.	4	1	2
	h).	What is community-based disaster management (CBDMM)?	4	1	2
	i).	What is GIS in disaster management?	5	1	2
	j).	What are early warning systems?	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain the concepts of disaster, hazard, risk, and vulnerability, and describe their relationship.	1	2	5
	b).	Explain the disaster profile of India and identify major high-risk zones with suitable examples.	1	2	5
OR					
3.	a).	Explain the classification of disasters into natural and man-made types with suitable examples.	1	2	5
	b).	Explain the relationship between human development and disasters.	1	2	5
UNIT-2					
4.	a).	Explain causes and effects of earthquakes and discuss basic earthquake engineering concepts.	2	2	5
	b).	Explain floods and urban flooding, and describe suitable flood mitigation measures.	2	2	5
OR					
5.	a).	Explain cyclone hazards and coastal vulnerability, and outline suitable mitigation strategies.	2	2	5
	b).	Explain man-made disasters such as structural failures and industrial hazards.	2	2	5

		UNIT-3			
6.	a).	Discuss the impacts of disasters on infrastructure, environment, and public health.	3	2	5
	b).	Explain structural and non-structural disaster mitigation strategies in civil engineering.	3	2	5
		OR			
7.	a).	Explain vulnerability and capacity assessment in disaster management.	3	2	5
	b).	Discuss global and national frameworks such as Sendai Framework and NDMA guidelines.	3	2	5
		UNIT-4			
8.	a).	Explain the disaster management cycle with neat diagram.	4	2	5
	b).	Discuss the role of community in disaster preparedness and resilience building.	4	2	5
		OR			
9.	a).	Explain emergency planning and post-disaster reconstruction	4	2	5
	b).	Discuss the importance of school safety programs and social capital in disaster recovery.	4	2	5
		UNIT-5			
10.	a).	Explain the role of GIS and Remote Sensing in disaster management.	5	2	5
	b).	Discuss early warning systems and disaster communication technologies.	5	2	5
		OR			
11.	a).	Explain the importance of IS codes in disaster-resilient infrastructure design.	5	2	5
	b).	Explain the role of indigenous knowledge and modern technologies in disaster risk reduction.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23CE4110					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
TRAFFIC ENGINEERING AND MANAGEMENT					
CIVIL ENGINEERING					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	State the relationship between speed, volume, and density.	1	2	2
	b).	What is the purpose of 85th percentile speed?	1	2	2
	c).	What is a parking inventory study?	2	1	2
	d).	State the objectives of Road Safety Audit.	2	2	2
	e).	What is Level of Service (LOS)?	3	1	2
	f).	Discuss any two roadway factors affecting LOS.	3	2	2
	g).	Mention the advantages of signalized intersections.	4	2	2
	h).	What is the purpose of IRC signal design method?	4	2	2
	i).	Define Transportation System Management (TSM)	5	2	2
	j).	Discuss any two parking management measures.	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Describe the objectives and applications of traffic volume studies.	1	2	5
	b).	Explain the methods used for spot speed studies with neat sketches.	1	2	5
OR					
3.	a).	Explain roadside interview method and home interview method used in O-D surveys.	1	2	5
	b).	Explain the basic traffic flow parameters namely speed, volume, and density with their interrelationships.	1	2	5
UNIT-2					
4.	a).	Discuss the objectives and importance of parking studies in urban traffic management.	2	2	5
	b).	Explain the causes of road accidents in detail.	2	2	5
OR					
5.	a).	Discuss different methods used for conducting parking surveys.	2	2	5
	b).	Explain the importance of accident studies in highway planning and traffic management.	2	2	5
UNIT-3					
6.	a).	Explain the basic concepts of traffic capacity and Level of Service	3	2	5

		(LOS).			
	b).	Explain the concept and significance of Passenger Car Unit (PCU) in capacity analysis.	3	2	5
		OR			
7.	a).	Explain the factors affecting highway capacity and Level of Service.	3	2	5
	b).	Explain the characteristics of urban highways and rural highways affecting traffic flow.	3	2	5
		UNIT-4			
8.	a).	Explain the different types of grade separated intersections with sketches.	4	2	5
	b).	Describe the procedure for determining optimum cycle length in traffic signal design as per IRC method.	4	2	5
		OR			
9.	a).	Explain the different types of intersections used in traffic engineering.	4	2	5
	b).	The average normal flow of traffic on cross roads A and B during design period are 400 and 250 PCU per hour; the saturation flow values on these roads are estimated as 1250 and 1000 PCU per hour respectively. The all-red time required for pedestrian crossing is 12 sec. Design two phase traffic signal with pedestrian crossing by Webster's method.	4	3	5
		UNIT-5			
10.	a).	Discuss various measures for improving vehicular flow in urban areas.	5	2	5
	b).	Discuss parking management strategies used in urban traffic control.	5	2	5
		OR			
11.	a).	Explain differential toll policies and their effectiveness in traffic management.			
	b).	Describe the role of signal improvement in reducing urban traffic congestion.	5	2	5
			5	2	5

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks